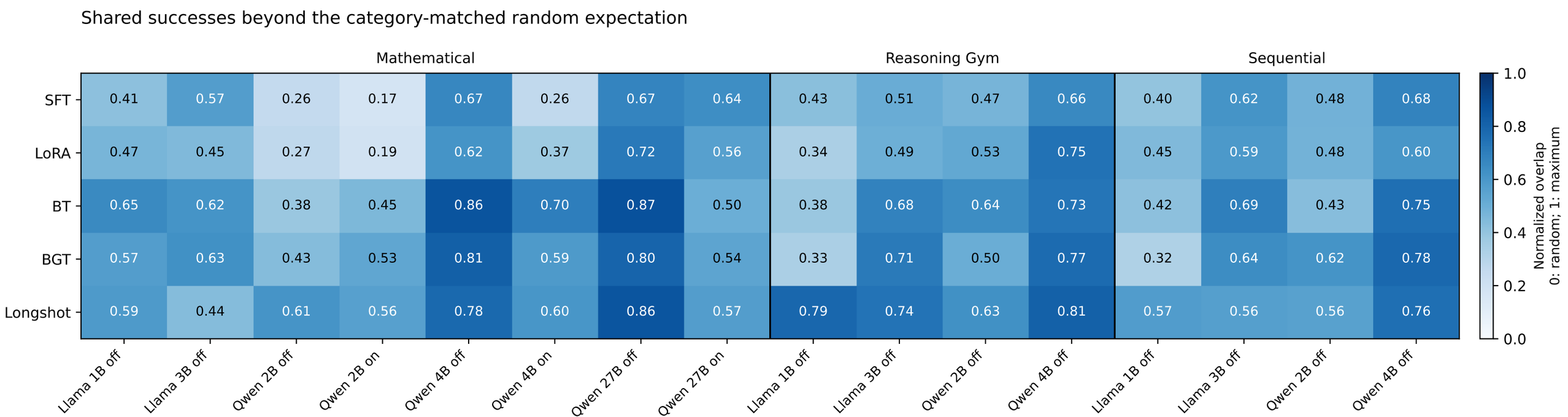
Shared successes beyond the category-matched random expectation
Mathematical
Reasoning Gym
Sequential
SFT
LoRA
BT
BGT
Longshot
0.41 0.57 0.26 0.17 0.67 0.26 0.67 0.64 0.43 0.51 0.47 0.66 0.40 0.62 0.48 0.68
0.47 0.45 0.27 0.19 0.62 0.37 0.72 0.56 0.34 0.49 0.53 0.75 0.45 0.59 0.48 0.60
0.65 0.62 0.38 0.45 0.86 0.70 0.87 0.50 0.38 0.68 0.64 0.73 0.42 0.69 0.43 0.75
0.57 0.63 0.43 0.53 0.81 0.59 0.80 0.54 0.33 0.71 0.50 0.77 0.32 0.64 0.62 0.78
0.59 0.44 0.61 0.56 0.78 0.60 0.86 0.57 0.79 0.74 0.63 0.81 0.57 0.56 0.56 0.76
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 2B on
Qwen 4B off
Qwen 4B on
Qwen 27B off
Qwen 27B on
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 4B off
Llama 1B off
Llama 3B off
Qwen 2B off
Qwen 4B off
1.0
0.8
0.6
0.4
0.2
0.0
Normalized overlap
0: random; 1: maximum